

Problem 1 – College Attendance Management System Implementation Report

Requirements Elicitation & Prioritization + Working Prototype

1. Selected Topic

College Attendance Management System. The prototype implements the core requirement areas identified in Problem 1: student attendance viewing, teacher attendance recording, biometric attendance as a prototype, management reporting, and a low-cost web-based structure.

2. Implementation Features

- Login page with role selection
- Admin attendance dashboard
- Student attendance table with present/absent status
- Biometric attendance prototype using roll-number verification
- Attendance reports with totals and percentage
- Flask + HTML/CSS implementation that can run locally

3. Implementation Screenshot

Figure 1 shows the prototype implementation screens prepared for the assignment.

Problem 1: Requirements Elicitation & Prioritization in a College Attendance System

Implementation (Sample) – College Attendance System

Scenario:

You are working as a Business Analyst for a college attendance management system. Students request app-based attendance checking, teachers request biometric integration, and management wants the solution to be low-cost and delivered quickly. Stakeholders have conflicting needs, and prioritization is required.

Deliverables

1. List of 4 well-defined requirements

ID	Requirement Description
R1	The system shall allow students to mark attendance using a mobile application.
R2	The system shall integrate biometric authentication for teachers to mark attendance.
R3	The system shall provide real-time attendance reports to management.
R4	The system shall be cost-effective and easy to deploy for the college.

2. MoSCoW Prioritization Table

Requirement ID	Requirement	Priority (MoSCoW)	Reason
R2	Biometric authentication for teachers	Must Have	Ensures accurate & secure attendance
R3	Real-time attendance reports	Should Have	Important for management decisions
R1	Mobile app attendance	Could Have	Enhances user convenience
R4	Low-cost and easy deployment	Won't Have (for now)	Budget optimization can be phased

3. Kano Model Classification

Requirement ID	Requirement	Kano Classification	Explanation
R2	Biometric authentication	Basic	Expected by teachers, absence causes dissatisfaction
R3	Real-time reports	Performance	Satisfaction increases as reporting improves
R1	Mobile app attendance	Excitement	Delights students but not expected initially
R4	Low-cost deployment	Basic	Expected by management

4. Stakeholder Conflict & Resolution

- **Conflict:** Students want mobile app attendance (R1) as top priority, while teachers want biometric integration (R2). Management is more concerned about cost (R4).

• Resolution Approach:

- ✓ Conduct stakeholder meeting to understand priorities.
- ✓ Use MoSCoW to agree on Must Have (R2) first.
- ✓ Plan phased release: Deliver R2 and R3 first, then R1 in next phase.
- ✓ Ensure cost optimization while delivering high-impact features.

1. Login Page

2. Dashboard (Admin View)

3. Mark Attendance (Biometric)

4. Reports Page

Figure 1: College Attendance System prototype – login, dashboard, biometric attendance and reports.

4. Technology Used

Frontend: HTML and CSS. Backend: Python Flask. Data: lightweight in-memory student records for demonstration. The design can later be extended with MySQL and a real biometric device/API.

5. How to Run

1. Extract the implementation ZIP. 2. Open a terminal in the extracted folder. 3. Run: pip install -r requirements.txt. 4. Run: python app.py. 5. Open http://127.0.0.1:5000/ in a browser.

6. Mapping to the Problem Requirements

Problem Requirement	Implementation
R1 – Student attendance checking	Dashboard displays attendance records and percentage.
R2 – Teacher attendance + biometric integration	Mark Attendance page demonstrates a biometric-verification workflow.
R3 – Management reports	Reports page shows total, present, absent and percentage values.
R4 – Low cost and quick delivery	Uses a lightweight Flask/HTML/CSS prototype with no expensive infrastructure.

7. Conclusion

The prototype demonstrates how the elicited requirements can be converted into a simple working software solution. The architecture is intentionally lightweight so that the first release can be developed quickly and expanded later with authentication, MySQL, mobile access, real biometric hardware, and role-based security.